SALES DATA ANALYSIS

Python Project — Source Code Documentation

A data analysis project using Pandas, NumPy, Matplotlib, and Seaborn to inspect, clean, summarize, and visualize sales data.

Project area	Exploratory Data Analysis (EDA)
Language	Python
Libraries	Pandas, NumPy, Matplotlib, Seaborn
Input dataset	sales.csv
Visualizations	Monthly trend, category sales, units sold, revenue heatmap

Before running the code

Install the required packages with `pip install pandas numpy matplotlib seaborn`. Update the CSV file path to match the location of your dataset. The CSV must contain the columns `Order Date`, `Item Type`, `Total Revenue`, and `Units Sold`.

Note: this PDF preserves the project logic and formats the code for reading. It does not include dataset output or generated chart images because the CSV file was not supplied.

Project Overview

1. Data loading and inspection

Loads sales.csv and prints sample rows, shape, columns, and data types.

2. Data quality checks

Checks missing values and duplicate records, removes duplicates, and fills selected missing values.

3. Sales analysis

Calculates revenue and quantity totals, average sales, high-value transactions, and product-level summaries.

4. Time-based analysis

Extracts year and month, aggregates daily/monthly revenue, and calculates month-over-month percentage change.

5. Visualization

Creates a monthly sales line chart, category sales bars, product quantity bars, and a monthly revenue heatmap.

Complete Python Source Code

The code is split into readable sections. Save it as `sales_analysis.py` and keep the dataset path correct before execution.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid", palette="muted")

# 1. Load Dataset
df = pd.read_csv("C:/Users/Lenovo/Downloads/sales.csv")

# Basic Info
print(df.head())
print(df.tail())
print(df.shape)
print(df.columns)
print(df.dtypes)

# Missing & Duplicate Check
print("Missing Values:\n", df.isnull().sum())
print("Duplicate Rows:", df.duplicated().sum())

# Clean Data
df = df.drop_duplicates()
df["Date"] = pd.to_datetime(df["Order Date"])
df["Item Type"] = df["Item Type"].fillna("Unknown")
df["Total Revenue"] = df["Total Revenue"].fillna(0)

# Quick Stats
print("Transactions:", len(df))
print("Products:", df["Item Type"].unique())
print("Categories:", df["Item Type"].nunique())
print("Total Revenue:", df["Total Revenue"].sum())

# High Sales
high_sales = df[df["Total Revenue"] > 10000]
print(high_sales)

# Filter
fruits_sales = df[
    (df["Item Type"] == "Fruits") &
    (df["Total Revenue"] > 1000)
]


print(fruits_sales)

# Sort by Revenue
sorted_sales = df.sort_values("Total Revenue", ascending=False)
print(sorted_sales.head())
```

```
# Total Revenue & Quantity
print("Total Sales:", df["Total Revenue"].sum())
print("Total Quantity:", df["Units Sold"].sum())

# Best Selling Product
product_sales = (
    df.groupby("Item Type")["Total Revenue"]
      .sum()
      .sort_values(ascending=False)
)
print(product_sales)

# Product Quantity
product_quantity = (
    df.groupby("Item Type")["Units Sold"]
      .sum()
      .sort_values(ascending=False)
)
print(product_quantity)

# Average Sales
print("Average Sale:", df["Total Revenue"].mean())

summary = df.groupby("Item Type").agg(
    Total_Sales=("Total Revenue", "sum"),
    Average_Sales=("Total Revenue", "mean"),
    Total_Quantity=("Units Sold", "sum"),
    Transactions=("Item Type", "count")
)
print(summary)

# ---------------- Time-Based Analysis ---------------- #

df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
df["Month_Name"] = df["Date"].dt.month_name()


# Daily Sales
daily_sales = df.groupby("Date")["Total Revenue"].sum()

# Monthly Sales
monthly_sales = df.groupby(["Year", "Month"])["Total Revenue"].sum()

# Best & Worst Month
print("Best Month:", monthly_sales.idxmax(), "Sales:", monthly_sales.max())
print("Worst Month:", monthly_sales.idxmin(), "Sales:", monthly_sales.min())

# Monthly Growth
monthly_growth = monthly_sales.pct_change() * 100
print("Monthly Growth:\n", monthly_growth)

# ---------------- Visualization with Seaborn ---------------- #

# 1. Monthly Sales
sns.lineplot(
    data=monthly_sales.reset_index(),
    x="Month", y="Total Revenue", hue="Year", marker="o"
)
plt.title("Monthly Sales Trend")
plt.show()

# 2. Category-wise Sales
sns.barplot(
    data=summary.reset_index(),
    x="Item Type", y="Total_Sales", palette="viridis"
)
plt.title("Total Sales by Category")
plt.xticks(rotation=45)
plt.show()

# 3. Product Quantity Distribution
sns.barplot(
    data=product_quantity.reset_index(),
    x="Item Type", y="Units Sold", palette="magma"
)
plt.title("Units Sold by Product")
plt.xticks(rotation=45)
plt.show()


# 4. Monthly Revenue Heatmap
monthly_matrix = df.pivot_table(
    index="Year", columns="Month_Name",
    values="Total Revenue", aggfunc="sum"
).fillna(0)

sns.heatmap(monthly_matrix, annot=True, fmt=".0f", cmap="coolwarm")
```

```
plt.title("Monthly Revenue Heatmap")
plt.show()
```